Robot Force Control on a Beating Heart

Rui Cortesão, *Member, IEEE*, and Michel Dominici, *Member, IEEE*

***Abstract*—Cardiac interventions have to deal with several difficulties, where heart motion is a major one. This paper addresses the heart motion compensation problem under contact, relying on robot force control techniques. A double active observer (AOB) architecture is proposed to tackle precise force control in the presence of heart motion. One AOB controls the desired interaction force, and the other one is responsible for compensating heart motion autonomously. Three-degrees of freedom (DOF) force tracking on top of a beating heart is presented, showing that with the proposed control architecture, it is possible to have force controlled interventions with root mean square errors below 0.56 N (based on raw data). The 4-DOF whole arm manipulator (WAM) from Barrett Technologies is the medical arm, while another 3-DOF robot carrying an *ex vivo* heart generates heart motion based on recorded biosignals.**

***Index Terms*—Autonomous motion compensation, beating heart, force control, Kalman active observer (AOB).**

I. Introduction

ROBOTIC-ASSISTED surgery is a challenging field of modern robotics, with over two decades of history. Although a lot of research has been done recently, long-standing problems such as precise force control, autonomous physiological motion compensation, and realistic haptic feedback (which is associated to force control) still have multiple scientific and technological issues that need to be addressed. In cardiac surgery, for instance, medical tasks are affected by disturbing forces due to breathing and heartbeats. Two well-known techniques address this issue. Passive mechanical stabilizers locally reduce heart motion, keeping errors in the range of 1–1.5 mm, which are still disturbing [2]. The other technique, more invasive, relies on stopping the heart and substituting it by an external machine. This solution entails more risks to patients, including inflammatory problems and long term cognitive loss. All these problems have motivated minimally invasive solutions, giving to the surgeon the impression of working on a motionless heart. This can be achieved by integrating robot control architectures that merge medical tasks with autonomous and synchronized heart motion compensation. Major works in this area use ultrasound [3], and vision [4], [5] to tackle the problem of heart motion, but little attention has been given to force control. A typical target application that can benefit from force control approaches is heart biopsy on a beating heart with controlled depth, since it requires accurate force control robust to motion disturbances. Other applications including delicate suturing and echocardiography on a beating heart also require accurate force control to boost performance. For instance, detailed ultrasound inspection of valves during beating heart surgery can help surgeons to assess and tune medical procedures in real time.

This paper presents the double active observer (AOB) architecture that tackles the force control problem under heart motion, and tests it with available robot and force sensor technologies. The architecture itself is not robot specific, and can be incorporated into other robotic platforms. Furthermore, force sensors can be eliminated based on techniques that estimate contact forces without explicit force sensing [6], [7].

This paper describes a three-dimensional (3-D) robot control architecture which has, for each Cartesian dimension, two independent AOBs for autonomous heart motion compensation and force tracking. This double AOB architecture does not rely on *a priori* knowledge of heart motion, and can be inserted into haptic telemanipulation schemes. A derivation of the double AOB architecture tested in the same robotic setup, which connects model predictive control and AOB (MPC-AOB), has recently been published with equivalent results [8], [9]. A major advantage of the double AOB architecture is its design procedure, since it has the same structure for both force control and motion compensation. In fact, only the stochastic design of an extended state (named active state) needs to be tuned for each AOB, leading to a simple tuning process. In the case of the MPC-AOB architecture, besides the stochastic design of one active state, there are other parameters of different nature that have to be tuned, creating more difficulties in the design process. These parameters are the length of the prediction horizon, the length of the control horizon, and two weights of a cost function. Hence, for the double AOB architecture there are only two parameters of the same nature to be tuned, while for the MPC-AOB architecture there are five parameters to be tuned, all with different nature.

The paper is structured as follows. The problem definition is addressed in Section II followed by related work analysis (Section III). The architecture for both force tracking and autonomous heart motion compensation is presented in Section IV. Section V discusses double AOB design. The overall computed

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R. Cortesão is with the Department of Electrical and Computer Engineering, Institute of Systems and Robotics, University of Coimbra, Coimbra 3030, Portugal (e-mail: cortesao@isr.uc.pt).

M. Dominici is with the Institute of Systems and Robotics, University of Coimbra, Coimbra 3030, Portugal (e-mail: zergspinal@gmail.com).

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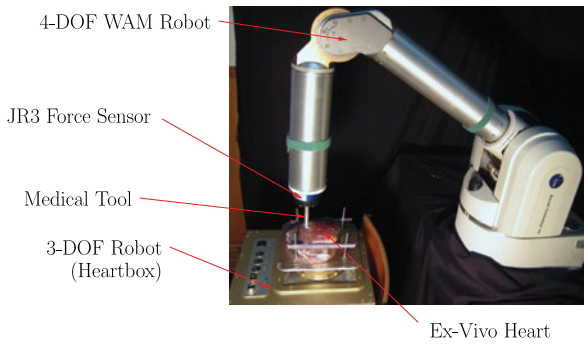


Fig. 1. Experimental setup. Four-DOF WAM robot with a JR3-67M25A3 force sensor at the end-effector in contact with an *ex vivo* heart. The heart is connected to a 3-DOF robot (called Heartbox), which generates 3-D heart motion.

torque technique is described in Section VI. Section VII shows 3-D experimental results and Section IX concludes the paper.

II. Problem Definition

This paper addresses the problem of controlling the force that a medical tool applies on a beating heart. Accurate 3-degrees of freedom (DOF) force tracking on a 3-DOF heart motion is required, even for time-varying reference forces. The experimental setup to tackle this problem is represented in Fig. 1. It is composed by a lightweight 4-DOF whole arm manipulator (WAM) carrying a force sensor and a medical tool (which is a thin aluminum rod). The heart motion is generated by a 3-DOF Cartesian robot (called Heartbox) based on recorded biosignals. The Heartbox reproduces 3-D motion with a precision of 0.1 mm.

III. Related Work

Heart motion tracking by either position, vision, or force sensing is a tough problem due to nonlinear deformations and complex lighting conditions. In [10], a position sensor to measure heart motion based on whisker sensors is proposed, although the integration in a control loop is not discussed. In [11], structured light is projected on organs, allowing relative instrument localization. Bachta *et al.* improved classical stabilizer solutions with piezo-electrical technology [12]. An H_∞ control architecture decreases, in one direction only, residual motion with vision feedback (the maximum residual motion is slightly above 0.6 mm for the controller that has no prior knowledge about the heart motion). This solution relies on a Cardiolock device that uses suction to grip the heart's surface, which may inflict tissue damage. Natural [13] or artificial [11] feature tracking and region tracking [14] are the major vision techniques to track heart motion. Recently [15], 1-DOF control techniques based on model predictive control has been proposed using medical imaging techniques, entailing position tracking errors of 8.1 mm. For this setup, no contact exists with the heart. Motion compensation using vision techniques referred to the medical contact point still remains an issue, since the contact point becomes nonvisible as the contact force is applied, deforming also the heart surface in the vicinity of contact. Moreover, estimating by visual inspection of tissue deformations heart motion requires precise modeling of the entire heart surface, which is also a difficult task. Force control architectures also have issues due to higher force sensor noise (even in free space). For haptic teleoperation schemes, robust performances (in free space and contact) can be achieved by generating reference forces based on position couplings [16]. Zarrouk *et al.* did preliminary studies on force control on a beating heart (1-DOF, only), where heart motion is modeled by a sinusoidal wave of frequency 0.125 Hz (i.e., only physiological motion due to respiration), being generated by an extra robot [17]. Kesner and Howe also worked on force control solutions for beating heart surgery, having pointed out catheter limitations due to backlash and friction (catheter solutions have a limited force range, though) [18], [19]. A summary of control approaches to compensate physiological motion is presented in [20], showing that force control has been little explored so far, as compared to position control.

IV. Double AOB Control Architecture

The double AOB architecture that tackles force control with autonomous physiological motion compensation is depicted in Fig. 2 [1]. The decoupled system plant $G(s)$ obtained by computed torque methods is the basis for the double AOB design.

A. Dynamic Modeling

Robot manipulator dynamics can be represented by

$$M(q)\ddot{q} + v(\dot{q}, q) + g(q) = \tau \tag{1}$$

where q and τ are the joint position and joint torque, respectively. $M(q)$, $v(\dot{q}, q)$, and $g(q)$ represent, respectively, the inertia matrix, Coriolis and centripetal torques, and the gravity term. Partially compensating robot dynamics, we get

$$M(q)\ddot{q} = \tau' \tag{2}$$

where[1]

$$\tau = \tau' + v(\dot{q}, q) + g(q). \tag{3}$$

In the operational space, (2) can be written as

$$\Lambda_t(q)\ddot{X}_t - \Lambda_t(q)\dot{J}_t(q)\dot{q} = F_t \tag{4}$$

with

$$\dot{X}_t = J_t(q)\dot{q} \tag{5}$$

$$\Lambda_t^{-1}(q) = J_t(q)M^{-1}(q)J_t^T(q) \tag{6}$$

and

$$\tau_t = J_t^T(q)F_t. \tag{7}$$

Λ_t, X_t, J_t, F_t, and τ_t are the operational space inertia matrix, Cartesian position, task space Jacobian (3×4), Cartesian force

[1]Throughout the paper we assume knowledge of robot dynamic and kinematic parameters with negligible errors. To simplify notation, dependencies are omitted whenever appropriate (e.g., $q(t) \equiv q$, $\Lambda_t(q) \equiv \Lambda_t$ and $F_t(t) \equiv F_t$). Additionally, all design parameters are based on SI units.

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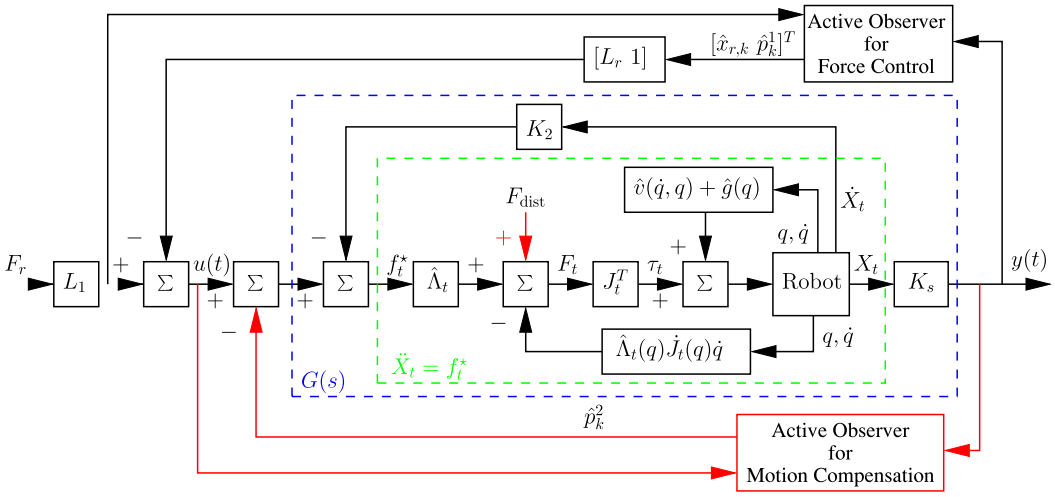


Fig. 2. Double AOB architecture for robot force control in the presence of heart motion [1]. The reference force F_r is tracked by the output force $y(t)$, based on desired tracking dynamics. The plant $G(s)$ includes $\ddot{X}_t = f_t^\star$, the real system stiffness K_s, a damping gain K_2 and the system delay T_d. $L_r = [L_1\ L_2\ L_3]$ is the state-feedback gain for the force loop. The physiological motion entails a disturbance force F_{dist} that is compensated by an extra AOB. $\hat{p}_k^1$ and $\hat{p}_k^2$ are the estimated disturbance states of the AOB for force control and motion compensation, respectively.

at the end-effector (where the task is defined), and task torque, respectively [21]. Note that τ' is given by

$$\tau' = \tau_t + \tau_{\text{null}} \tag{8}$$

where τ_{null} represents torques that do not affect the task. In this paper, the null space is not explored, therefore, τ_{null} is set to zero.

B. Feedback Linearization

Without disturbance forces,[2] F_t is equal to the commanded force $F_{c,t}$, which is computed by

$$F_{c,t} = -\Lambda_t(q)\dot{J}_t(q)\dot{q} + \Lambda_t(q) f_t^\star \tag{9}$$

to achieve the desired decoupled system

$$\ddot{X}_t = f_t^\star. \tag{10}$$

$f_t^\star$ is a control input parameter. Introducing the damping term K_2 and considering also the system time delay T_d (mainly due to signal processing), and a spring contact[3] represented by the environment stiffness K_s, the linear system plant $G(s)$ for each Cartesian dimension is given by (see Fig. 2)

$$G(s) = \frac{K_s\, e^{-sT_d}}{s(s + K_2\, e^{-sT_d})}. \tag{11}$$

Since T_d is small, the plant used in control design is approximated by

$$G(s) \approx \frac{K_s\, e^{-sT_d}}{s(s + K_2)} \tag{12}$$

which is represented in time domain by

$$\ddot{y}(t) + K_2\dot{y}(t) = K_s u(t - T_d), \tag{13}$$

where $y(t)$ is the measured force at the robot end-effector and $u(t)$ is the plant input (acceleration). Defining the state variables $x_1(t) = y(t)$ and $x_2(t) = \dot{y}(t)$, (13) is written as

$$\begin{bmatrix}\dot{x}_1(t)\\ \dot{x}_2(t)\end{bmatrix} = \begin{bmatrix}0 & 1\\ 0 & -K_2\end{bmatrix}\begin{bmatrix}x_1(t)\\ x_2(t)\end{bmatrix} + \begin{bmatrix}0\\ K_s\end{bmatrix} u(t - T_{\text{d}}). \tag{14}$$

The discrete representation of (14) is of form [16], [23]

$$\begin{cases} x_{r,k} = \Phi_r x_{r,k-1} + \Gamma_r u_{k-1} \\ y_k = C_r x_{r,k} \end{cases}. \tag{15}$$

$x_{r,k}$ is a 3-D vector (reduced state), since $T_d = h$, where h is the sampling time. $x_{r,k}$ is composed by the end-effector force (first state), its derivative (second state), and u_{k-1} (third state) due to time delay. The state-space plant description (14) and (15) will be used by both AOBs to control the system using state augmentation techniques.

V. Double AOB Design

This section provides the technical background associated to the double AOB architecture represented in Fig. 2. The AOB extends the Kalman filter formulation to achieve model reference adaptive control.

A. AOB Concept

The AOB reformulates the Kalman filter based on the following.

1) A desired closed loop representation, corresponding to the reference model.
2) An active state p_k (extra state), which describes an equivalent disturbance referred to the system input. Its estimate $\hat{p}_k$ is responsible for compensating actions. p_k

[2]In the real scenario, disturbance forces are mainly due to cardiac motion. Nevertheless, a disturbance free model has to be known for model reference adaptive control.

[3]A spring model has been chosen, although the analysis can be extended to other contact models. See [22] for a literature review on modeling tool-tissue interactions.

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is mathematically represented by

$$p_k = \sum_{j=1}^{N} (-1)^{j+1} \frac{N!}{j!(N-j)!} p_{k-j} + {}^{N-1}\xi_{p_k}. \quad (16)$$

Equation (16) illustrates the random distribution of the Nth-order derivative of p_k. The variable ${}^{N-1}\xi_{p_k}$ is Gaussian with zero-mean.

3) The stochastic design for the AOB context.

B. AOB-N Algorithm

Equation (16) can be written in state space as

$${}^{N}p_k = \Phi_{2,2}\ {}^{N}p_{k-1} + \ \xi_{{}^{N}p_k} \quad (17)$$

with

$${}^{N}p_k = \begin{bmatrix} p_{k-(N-1)} & p_{k-(N-2)} & \cdots & p_{k-1} & p_k \end{bmatrix}^T \quad (18)$$

$$\Phi_{2,2} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ a_N & a_{N-1} & a_{N-2} & \cdots & a_1 \end{bmatrix} \quad (19)$$

$$a_i = (-1)^{i+1} \frac{N!}{i!(N-i)!},\ i = 1, \ldots, N \quad (20)$$

and

$$\xi_{{}^{N}p_k} = \begin{bmatrix} 0 & 0 & \cdots & 0 & {}^{N-1}\xi_{p_k} \end{bmatrix}. \quad (21)$$

Incorporating p_k as an extra plant input, and adding stochastic uncertainty to $x_{r,k}$ through the zero-mean Gaussian variable $\xi_{x_{r,k}}$, the overall system can be represented by

$$\begin{bmatrix} x_{r,k} \\ {}^{N}p_k \end{bmatrix} = \begin{bmatrix} \Phi_r & \Phi_{1,2} \\ 0 & \Phi_{2,2} \end{bmatrix} \begin{bmatrix} x_{r,k-1} \\ {}^{N}p_{k-1} \end{bmatrix} + \begin{bmatrix} \Gamma_r \\ 0 \end{bmatrix} u'_{k-1} + \begin{bmatrix} \xi_{x_{r,k}} \\ \xi_{{}^{N}p_k} \end{bmatrix} \quad (22)$$

where

$$\Phi_{1,2} = \begin{bmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{bmatrix} \quad (23)$$

and

$$u'_{k-1} = r_{k-1} - \begin{bmatrix} L_r & 0 & \cdots & 1 \end{bmatrix} \begin{bmatrix} x_{r,k-1} \\ {}^{N}\hat{p}_{k-1} \end{bmatrix}. \quad (24)$$

The state feedback gain L_r of form

$$L_r = [\, L_1\ L_2 \cdots\ L_M \,] \quad (25)$$

can be obtained for a desired closed loop dynamics. Canceling the influence of p_k by perfect estimation and compensation entails a decoupled *a priori* state estimate X_e^-, i.e.,

$$\begin{bmatrix} \hat{x}_{r,k} \\ {}^{N}\hat{p}_k \end{bmatrix} = X_e^- + K_k \left\{ y_k - C X_e^- \right\} \quad (26)$$

with

$$X_e^- = \begin{bmatrix} \Phi_{L,n} & 0 \\ 0 & \Phi_{2,2} \end{bmatrix} \begin{bmatrix} \hat{x}_{r,k-1} \\ {}^{N}\hat{p}_{k-1} \end{bmatrix} + \begin{bmatrix} \Gamma_{r,n} \\ 0 \end{bmatrix} r_{k-1} \quad (27)$$

and

$$\Phi_{L,n} = \Phi_{r,n} - \Gamma_{r,n} L_r. \quad (28)$$

The Kalman gain K_k is given by

$$K_k = P_{1k}\, C^T \left[C\, P_{1k}\, C^T + R_k \right]^{-1} \quad (29)$$

with

$$P_{1k} = \Phi_n\, P_{k-1}\, \Phi_n^T + Q_k \quad (30)$$

and

$$P_k = P_{1k} - K_k\, C\, P_{1k}. \quad (31)$$

Φ_n is the extended open loop matrix

$$\Phi_n = \begin{bmatrix} \Phi_{r,n} & \Phi_{1,2} \\ 0 & \Phi_{2,2} \end{bmatrix} \quad (32)$$

where $\Phi_{r,n}$ represents the numerical representation of Φ_r used in control design. $\Phi_{r,n}$ incorporates an estimate of K_s, $\hat{K}_s$, corresponding to the desired contact model. The measurement matrix C is set to

$$C = \begin{bmatrix} C_r & 0 \end{bmatrix}. \quad (33)$$

K_k is a function of both the system uncertainty matrix Q_k and the measurement uncertainty matrix R_k. In the AOB context, Q_k is represented by

$$Q_k = \begin{bmatrix} Q_{x_{r,k}} & 0 \\ 0 & Q_{{}^{N}p_k} \end{bmatrix}. \quad (34)$$

Merging uncertainties into ${}^{N}p_k$ makes the other system states more deterministic (which correspond the closed-loop reference model), affecting Q_k design. Only the relative gains of Q_k and R_k have effect on the Kalman gain [16].

C. AOB for Force Control

Force tracking dynamics between F_r and $y(t)$ are imposed by the "Active Observer for Force Control" (see Fig. 2). Since the state feedback gain L_r is a function of K_2, and the force derivative is the noisiest signal, K_2 is tuned to reduce noise in the control loop. Bigger K_2 entails a smaller force derivative gain L_2. The desired contact stiffness $\hat{K}_s$ should not be too different from the real contact stiffness K_s to enhance force tracking performance.

1) Stochastic Design: A first-order AOB has been chosen for force control, guaranteeing robustness to stiffness errors [16]. Therefore, Q_k is a 4×4 diagonal matrix, where $Q_{x_{r,k}}$ is a 3×3 matrix whose values are much smaller than $Q_{{}^{N}p_k}$. R_k is set to one. Therefore, $Q_{{}^{N}p_k}$ is the only matrix that needs to be experimentally tuned (for the double AOB architecture, it is just one parameter).

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2) Limitations: Control and stochastic designs are tuned for force tracking, entailing poor rejection capabilities under persistent disturbance forces due to physiological motion. Increasing $Q_{N\,p_k} \equiv Q_{p_k^1}$ or the AOB order would improve rejection ratios, having however undesired implications in the state estimate $\hat{x}_{r,k}$. This fact motivated the addition of another AOB to cope with physiological motion.

D. AOB for Motion Compensation

1) Control Issues: The "AOB for Motion Compensation" only takes into account the double integrator plant, system time delay and $\hat{K}_s$. No state feedback control exists except the active state feedback of $\hat{p}_k^2$. Since physiological motion is well described by second-order functions, a second-order AOB has been designed. Therefore, the p_k^2 dynamics is represented by two states. $\hat{p}_k^2$ can be seen as an acceleration input to autonomously compensate physiological motion.

2) Stochastic Design: $Q_{x_{r,k}}$ and R_k have the same design of the "Active Observer for Force Control." The other parameter is

$$Q_{N\,p_k} = \begin{bmatrix} 0 & 0 \\ 0 & Q_{p_k^2} \end{bmatrix}. \tag{35}$$

It should be noted that each AOB has its own $x_{r,k}$ and p_k estimations. Again, $Q_{N\,p_k}$ is the only matrix that needs to experimentally tuned (for the double AOB architecture, it is just one parameter).

3) Limitations: A conservative value of $Q_{p_k^2}$ (i.e., small value) improves robustness, although it degrades motion compensation capabilities.

VI. Commanded Torque

From (3), (7), and (9), the control torque τ_c is given by

$$\tau_c = v(\dot{q}, q) + g(q) + J_t^T(q)\Lambda_t(q)\left\{f_t^\star - \dot{J}_t(q)\dot{q}\right\} \tag{36}$$

$f_t^\star$ is a 3-D vector for Cartesian control (the same applies for F_r and $\dot{X}_t$). Its components $f_{t,i}^\star$ are given by

$$f_{t,i}^\star = L_1 F_{r,i} - [L_r\ 1][\hat{x}_{r,k}\ \hat{p}_k^1]^T - \hat{p}_k^2 - K_2\dot{X}_{t,i}. \tag{37}$$

The estimates $\hat{x}_{r,k}$, $\hat{p}_k^1$, and $\hat{p}_k^2$ in (37) are performed by the double AOB architecture associated to each task dimension. $f_{t,i}^\star$ is an acceleration input for each Cartesian dimension.

VII. Experiments

To assess the double AOB architecture, 3-D breathing and heartbeat signals are reproduced by the Heartbox, based on physiological data obtained from *in vivo* experiments on a pig's heart [24]. The WAM robot applies 3-D forces on the moving heart, in spite of force disturbances associated to breathing and cardiac motions. A desired time varying force along Z keeping all lateral forces (along X and Y) equal to zero is the chosen scenario to assess force control performance on a beating heart. Thus, the desired force is 0 N for X and Y, and time varying for Z, including constants, ramps and an up-chirp signal from 0.2 to 1 Hz, representing an hypothetical medical force.

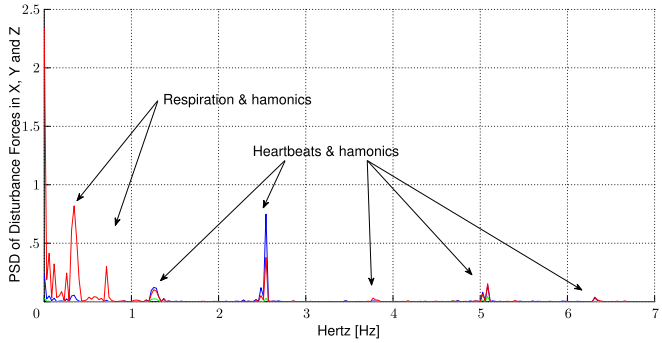


Fig. 3. PSD of estimated disturbance forces referred to the contact point due to breathing and heartbeat motions. X, Y, and Z signals are shown by blue, green, and red plots, respectively.

This desired force is a synthetic generated signal, whose power spectral density (PSD) is within human arm bandwidth.[4] The overall double AOB control architecture is implemented on a 2.1 GHz Intel Core 2 processor running Xenomai-Linux. The control sampling time is set to $h = 1$ ms.

A. AOB Design for Force Control

For a critically damped force tracking with time constant equal to 3 ms, $K_2 = 10$ and $\hat{K}_s = 900$ N/m, we get

$$L_r = \begin{bmatrix} 161.5 & 1.189 & 0.557 \end{bmatrix}. \tag{38}$$

$R_k = 1$ and Q_k is set to

$$Q_{x_{r,k}} = \begin{bmatrix} 10^{-12} & 0 & 0 \\ 0 & 10^{-12} & 0 \\ 0 & 0 & 10^{-12} \end{bmatrix} \tag{39}$$

and

$$Q_{1\,p_k} = 0.5 \tag{40}$$

entailing the steady state AOB gain

$$K_{k,f} = \begin{bmatrix} 0.1236 & 8.145 & 0.662 & 0.662 \end{bmatrix}^T. \tag{41}$$

B. AOB Design for Motion Compensation

For the second AOB, $R_k = 1$ and Q_k is a 5×5 matrix, where $Q_{x_{r,k}}$ is equal to (39) and $Q_{2\,p_k}$ is set to

$$Q_{2\,p_k} = \begin{bmatrix} 0 & 0 \\ 0 & 5.0 \end{bmatrix}. \tag{42}$$

This stochastic design with $\hat{K}_s = 900$ N/m generates the steady-state AOB gain

$$K_{k,m} = \begin{bmatrix} 0.3738 & 87.08 & 26.25 & 26.25 & 28.02 \end{bmatrix}^T. \tag{43}$$

[4]The desired forces do not come from a real medical task, however the magnitude and type of involved signals are rich enough to consider our force control architecture as a strong candidate for clinical assessment. Before reaching this stage, research on haptic telemanipulation architectures incorporating the specificities of the force control approach needs to be carried out.

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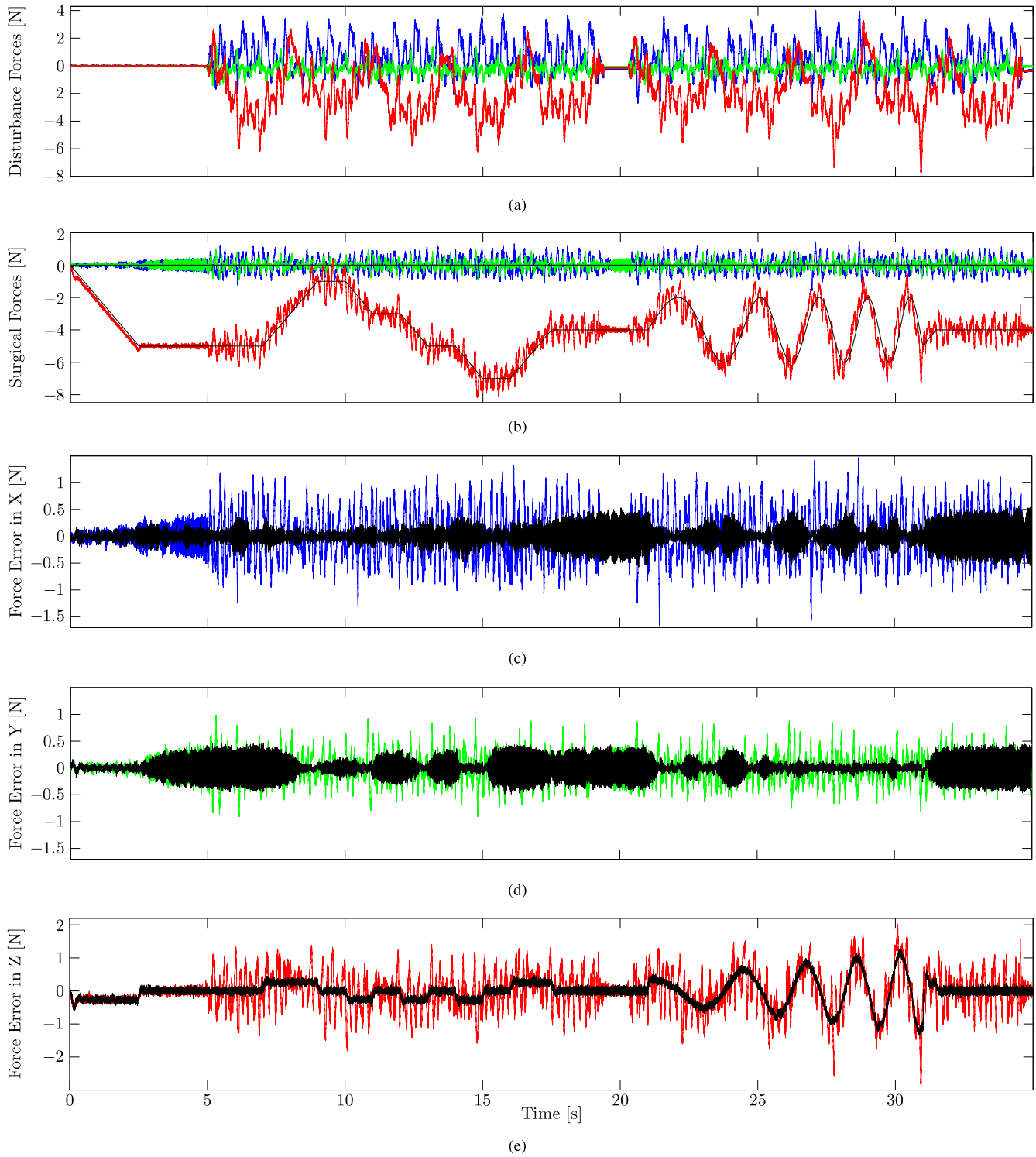


Fig. 4. Double AOB force control in the presence of disturbance forces due to breathing and heartbeats. Experimental results. X, Y, and Z signals are shown by blue, green, and red plots, respectively. (a) Estimated disturbance forces referred to the contact point. (b) The 3-D reference force F_r is represented in black, while measured forces are in blue, green, and red. (c), (d), and (e) Force errors with the Heartbox turned ON (colored plots) and OFF (black plots).

C. Force Control on a Beating Heart

For analysis purposes, force disturbances due to breathing and cardiac motions seen by the WAM robot are estimated as follows. Two different experiments record WAM robot Cartesian positions and residual force measurements. In the first experiment, the Heartbox is turned OFF to get the motion due to the reference force alone. In the second experiment, the Heartbox is turned ON to get the motion due to the reference force plus motion compensation actions. WAM robot displacements with the Heartbox turned ON are subtracted from those with the

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TABLE I
RMSE OF THE FORCE FOR A SPECIFIC SET OF TIME INTERVALS

	Compression	Constant force	Releasing	Harmonic excitation
Time (s)	[10–11]; [12–13]; [14–15]	[5.1–7]; [9–10]; [11–12]; [13–14]; [15–16]; [17.5–19]; [31.5–34.3]	[16–17.5]; [31–31.46]	[21–30.9]
X-axis (N)	**0.49**	**0.48**	**0.46**	**0.48**
	0.15	0.20	0.17	0.16
Y-axis (N)	**0.29**	**0.27**	**0.26**	**0.25**
	0.09	0.22	0.23	0.09
Z-axis (N)	**0.61**	**0.51**	**0.50**	**0.74**
	0.27	0.07	0.29	0.56

Bold values are for Heartbox turned ON and nonbold values are for Heartbox turned OFF.

Heartbox turned OFF. These displacements multiplied by the estimated heart stiffness[5] 810 N/m are mapped into force data. To these force data we add the force errors from the experiment with Heartbox turned ON, leading to an estimate of force disturbances referred to the contact point. Fig. 3 shows the PSD of such disturbances and Fig. 4(a) presents the corresponding time plots.[6] From Fig. 3, it can be inferred that the first two peaks around 0.34 and 0.72 Hz are due to respiration, corresponding to 20 breathing cycles per minute. The last five peaks around 1.25, 2.54, 3.77, 5.09, and 6.31 Hz are due to heartbeats, which correspond to 75 heartbeats per minute. Fig. 4 shows force control results. The Heartbox is turned ON around 5 s and goes up to around 20 s, where there is a 1 s pause. After the pause, the motion is repeated. Fig 4(b) shows the 3-D reference force F_r in black, while measured forces are depicted in blue, green, and red. Force errors are shown in Fig. 4(c)–(e), being typically below 1 N for X and Y and below 1.6 N for Z, with global root mean square (rms) errors below 0.56 N for all axis, corresponding to uncompensated motion in the order of 0.69 mm. It should be stressed again that these results are based on raw data. When the force sensor moves back and forward during the heart motion compensation task, it feels inertial forces due to its mass and the mass of the tool attached to it, amplifying force sensor noise. This effect is stronger in Z and X directions, since they carry stronger heart motion activity. For instance, around 5 s and around 20 s, where reference forces along X, Y, and Z are constant, the noise pattern is affected [see Fig. 4(b)] when the Heartbox is turned ON [see Fig. 4(a)]. Black plots of the force error show the double AOB control performance when the Heartbox is turned OFF all the time. As expected, force tracking is degraded as signal dynamics increases. Table I shows partial root mean square errors (RMSE) of the force for several situations during the time interval 5–34.3 s. Overall the errors are small, being higher when the Heartbox is ON (as expected). The Z direction has the highest errors due to both time varying reference force and significant heart motion along Z. Moreover, looking to the PSD in Fig. 3, we can infer that cardiac activity along X is higher than along Y, entailing higher RMSE in X (w.r.t. Y). When the Heartbox is OFF, RMSE in X and Y are relatively high for constant forces in Z. This is due to the almost static "valley" of cardiac tissues that surrounds the tool, generating persistent disturbance forces in X and Y, as the controller tries to reach 0 N for all lateral forces.

[5]The estimation is done offline from force and position data using Hooke's law. For a step force from 1 to 5 N, we measured the corresponding displacement to have a rough estimate of the heart stiffness.

[6]All data represented in Figs. 3 and 4 are obtained from robot encoder measurements and force sensor measurements without filtering (i.e., raw data).

VIII. DISCUSSION

In the past, stand-alone AOB architectures have been successfully used to tackle multiple force control problems in static environments, only (e.g., [16], [25], [26]). The present paper shows that for a moving heart it is also possible to have accurate force control using a double AOB architecture. The double AOB architecture is an elegant extension of stand-alone AOB architectures, where two AOBs, side by side, independently address force control and environmental motion compensation. The fundamental problem of 3-D force control on a beating heart is solved within the AOB framework, requiring straightforward and simple tuning procedures. Only taking into account raw data, the maximum amplitude errors are typically below 1 N for X and Y and below 1.6 N for Z, with global rms errors below 0.56 N (worst case scenario), corresponding to uncompensated motion in the order of 0.69 mm. These results have been attained for 3-DOF scenarios without knowing heart motion in advance, paving the way to force controlled interventions on a beating heart.

IX. CONCLUSION

We presented a double AOB architecture for robot force control in the presence of heart motion. Two separate AOBs are designed for force control and autonomous heart motion compensation. The architecture relies on computed torque, linked to robot task space techniques, discrete state space estimation, and control. A realistic beating heart scenario has been used in the experiments without living beings in the workspace, which consists of a 3-DOF Heartbox robot equipped with an *ex vivo* heart that generates 3-D motion based on recorded biosignals from a pig heart. Additionally, a 4-DOF WAM robot manipulator carrying a force sensor and a medical tool applies controlled forces on the moving heart. Experimental results show that the 3-D force control problem on a beating heart is solved within the AOB framework, keeping global rms force errors small, which correspond to submillimeter accuracy.

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Rui Cortesão (S'01–M'04) received the B.Sc. ("Licenciatura") degree in electrical engineering, the M.Sc. degree in systems and automation, and the Ph.D. degree in control and instrumentation from the University of Coimbra, Coimbra, Portugal, in 1994, 1997, and 2003, respectively.

He was a Visiting Researcher at the German Aerospace Center - DLR (1998–2003), where he did his Ph.D. work; Stanford University, Stanford, CA, USA, (2002); LIRMM-CNRS (2004, 2006, and 2008); and Barrett Technologies (2007), having worked on surgical robotics, haptic tele-manipulation, compliant motion control, data fusion, and human-robot skill transfer. He is the Head of the Medical Robotics Group with the Institute of Systems and Robotics, University of Coimbra, and an Assistant Professor with the University of Coimbra.



Michel Dominici (M'12) received the B.Sc. degree in electrical engineering, and the M.Sc. degree in system and automation (major in automation) from the University of Montpellier II, Montpellier, France, in 2003 and 2005, respectively, and the Ph.D. degree in robotics and automation from the University of Coimbra, Coimbra, Portugal, in 2016.

From 2006 to 2009, he worked as a Researcher on surgical robotics projects at LIRMM-CNRS, France. He was a Researcher at the Institute of Systems and Robotics, University of Coimbra (2010–2016).

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